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TECHNICAL SKILLS

CAD & Mechanical Design	Fusion 360 SolidWorks AutoCAD Blender COMSOL CAM
Embedded Systems & Programming	Python C++ C# Arduino Raspberry Pi ESP32 Sensor Integration API & Tool development Computational Geometry
Prototyping, Fabrication & Simulation	Simulation Automation DFM for CNC/Waterjet/3D print/MEMS Data Processing & Visualization Hand & Power Tools Soldering

EDUCATION

University of British Columbia

Bachelor of Applied Science - Integrated Engineering, Mechatronics Focus

Expected: Apr. 2029

ENGINEERING OR DESIGN STUDENT TEAMS

- Canada-Wide Science Fair, Youth Science Canada

Mar. 2024 – Jun. 2024

Finalist

 - Designed and prototyped **Leaf Walker**, an innovative forestry chassis concept using a unique virtual pivot steering system to minimize soil damage and improve maneuverability over conventional articulated designs, resulting in 32% less shear area.
 - Awarded Best in Class and the Transport Award at the regional level before advancing to nationals, delivering technical presentations to technical and non-technical audiences.
- Suspension Team Member, UBC BAJA

Sep. 2025 – Present

 - Designed, validated and manufactured suspension components for a BAJA vehicle to compete in future competitions through stringent safety factor analysis.
 - Manufactured and assembled parts as a team and met project deadlines.
 - Developing testing methods and software for design validation and feedback using BeamNG.tech simulation utilizing previous algorithmic geometry experience to allow easy changes between vehicle generations, as well as parameter-based geometry for suspension and the gearbox.

TECHNICAL EXPERIENCE AND PROJECTS

- Bosch Research Stuttgart, Germany

May 2026 – Present

Audio MEMS Development & Concept Engineering

 - Applied macro-scale mechanical systems expertise to MEMS geometry optimization, identifying cross-domain design improvements within established development workflows.
 - Developed an automated simulation and analysis pipeline combining parameterized geometry, compute-cluster execution, and custom post-processing to identify optimal design regions and performance boundaries.
 - Developing a functional macro-scale demonstrator of a Family of MEMS devices to communicate dynamic behavior and engineering concepts to technical and business stakeholders.
- Gesture Link Capstone Project

Feb. 2026 – Apr. 2026

 - Proposed and led development of a camera-free wearable finger-tracking prototype using a pulsed electromagnetic ring, tri-axial sensor array, and embedded acquisition hardware.
 - Led a multidisciplinary team of five through system development, technical direction, and milestone planning while mentoring teammates on fundamental concepts.
 - Developed and validated an embedded data pipeline integrating sensor multiplexing, Arduino acquisition, and signal processing, achieving strong correlation between simulated and measured magnetic fields.